

ADITYA SUNIL TAWDE

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EDUCATION

Bachelor of Technology — Artificial Intelligence & Data Science

Jawaharlal Nehru Engineering College (JNEC), MGM University, Chh. Sambhajnagar | Expected: May 2028

TECHNICAL SKILLS

Languages: Python, JavaScript, C/C++, HTML/CSS

Frameworks: FastAPI, Django, Flask, React, NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Transformers, OpenCV, Selenium

AI/ML: LLMs (LlamaCpp/Groq/Gemini), RAG (BM25 + Dense), sentence-transformers, GGUF quantization, Whisper, llama-cpp-python

Databases: PostgreSQL, SQLite, Redis, Qdrant (vector DB), SQLAlchemy, Alembic

Tools: Git/GitHub, Docker, VS Code, Plotly, Matplotlib, Vite, Firebase

PROJECTS

Amadeus AI v4.0.0 — Python, FastAPI, SQLAlchemy, LlamaCpp, Groq, Gemini, Redis, Qdrant, PostgreSQL, Docker, JWT

- **Autonomous AI operating layer** built on Clean Architecture — transport → application → core → infrastructure — with a plan-driven cognitive runtime and persistent execution graph (Plan → PlanStep → Observation → Reflection).
- **LLM routing chain:** LlamaCpp (local GGUF) → Groq (Llama 3.3 70B) → Gemini 2.5 Flash, with Redis-backed daily quota tracking; LOCAL_ONLY_MODE for 100% on-device privacy.
- **70+ sandboxed tools** across system control, network, filesystem, and productivity categories; hot-pluggable plugin system, HITL approval gate for destructive actions, JWT auth, and prompt-injection resistance.
- **Hybrid RAG engine:** BM25 + dense vector search (Qdrant) with metadata-enriched chunking; voice pipeline via Whisper STT → LLM → Edge TTS with real-time SSE streaming.

Arth-Neeti — Financial Literacy RPG — Django 5, React 18, Firebase Auth, Google Gemini, Groq, SQLite

- Turn-based financial simulation game covering 12 months of professional life with real-time stock market simulation, credit scoring (300-900), and dynamic loan interest calculation.
- **7 distinct AI advisor personas** (Friendly, Strict, Sassy, Harshad, Vasooli, etc.) built on Gemini, triggered by in-game events; proactive advice fires contextually (e.g., debt collector appears on missed EMI).
- Trilingual support (English, Hindi, Marathi) via google-genai and deep-translator; post-game analytics assigns financial persona ("The Warren Buffett", "The FOMO Victim") with Gemini-generated Markdown report.

Ledger — AI-Native Personal Finance Platform — FastAPI, React, PostgreSQL, PaddleOCR, Groq, Gemini, Cerebras

- **Multi-provider LLM fallback router:** Groq → Gemini → Cerebras → Mistral, with receipt OCR (PaddleOCR), AI auto-categorization, and SSE-streamed advisor chat.
- 35+ REST endpoints covering transactions, budgets, investments (stocks/MF/crypto/FD), GST computation, and Tally XML export; credit health scoring and community benchmarks.
- Deployed on Vercel + Render free tier; solved Render cold-start via staggered GitHub Actions + Vercel cron ping layers.

Amadeus-chat — Local LLM CLI — Python, llama-cpp-python, sentence-transformers, BM25, cross-encoder reranking, Rich

- **100% on-device inference** using GGUF-quantized models (Qwen3.5-2B Q4_K_M); runs on 4 GB RAM / CPU-only hardware with no cloud dependency.

- Hybrid RAG: BM25 keyword + dense semantic retrieval fused via Reciprocal Rank Fusion (RRF), cross-encoder re-ranking, and MD5 deduplication; supports PDF, MD, CSV, JSON, HTML, PY.
- Sliding window + LLM-powered memory compression with token budget guard (warn at 80%, block at 95%); whitelisted safe shell tool execution via psutil.

DataScraperviz — Market Data Intelligence — Python, Alpha Vantage API, Selenium, Pandas, Matplotlib, Seaborn, Plotly

- Scraped real-time stock market data (price, market cap, P/E ratio, volume) and LinkedIn job listings using Selenium; persisted to structured CSV for downstream analysis.
- Generated interactive visualizations: market cap distribution by sector, top-10 company rankings, and stock price vs. volume scatter plots using Plotly.

LibraryPro — Desktop Library Management System — Java 11, SQLite, Apache POI, FlatLaf, Swing

- MVC-architected library system with database-backed auth, bulk CSV/Excel import (duplicate skipping), 14-day loan tracking, and ₹1/day overdue fine calculation.
- Role-based access (admin/librarian), live dashboard KPIs, full-text catalog search, and cross-platform build pipeline (Windows .exe / macOS .dmg / Linux .AppImage).

CERTIFICATIONS

Gen AI Foundational Models for NLP & Language Understanding — IBM / Coursera (Mar 2026)

Generative AI and LLMs: Architecture and Data Preparation — IBM / Coursera (Feb 2026)

Introduction to Deep Learning & Neural Networks with Keras — IBM / Coursera (Feb 2026)

Machine Learning with Python — IBM / Coursera (Jan 2026)

Building Generative AI-Powered Applications with Python — IBM / Coursera (Oct 2025)

Developing AI Applications with Python and Flask — IBM / Coursera (Oct 2025)

Machine Learning Foundations: A Case Study Approach — University of Washington / Coursera (Jul 2025)

Programming with JavaScript — Meta / Coursera (Nov 2025)

ACHIEVEMENTS & ACTIVITIES

- Pitched **Printhub** at E-Summit 2026 Udyam Startup Pitching Competition, JNEC MGM University (Feb 2026).
- Participated in Concept2Code and Tech-Escape Room at Swayambhu 2026 Technical Fest, JNEC MGM University (Feb 2026).
- Shipped a functional application during Dreamflow Buildathon 2025, demonstrating proficiency in the FlutterFlow ecosystem (Certificate ID: 2026H2S02DFLOW-P00078).

INTERESTS

Agentic AI Systems • LLM Engineering • Backend Architecture • Automation • Japanese Language Learning